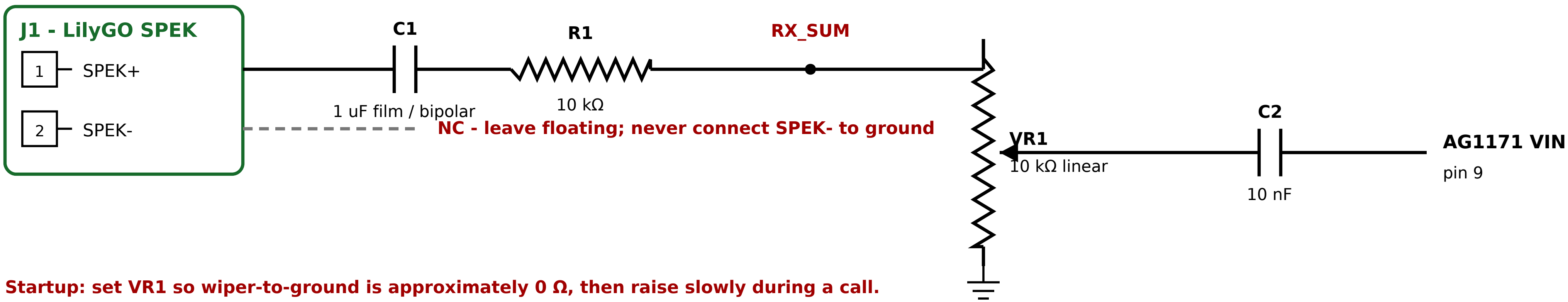
ESP-CellCore Passive Audio Interface Board - Rev A.1

Sheet 1 - Receive audio and GPIO36 tone injection

PROTOTYPE - BENCH VALIDATION REQUIRED

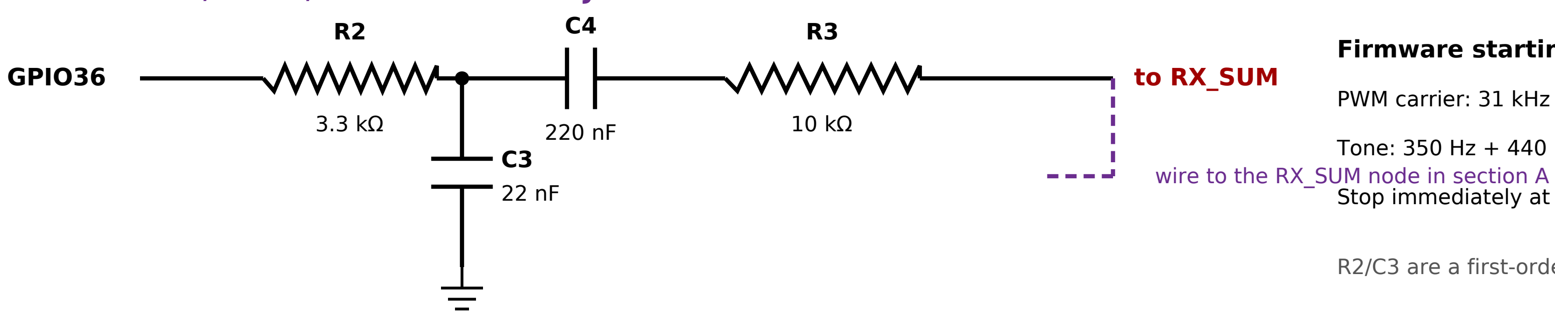
A. RECEIVE: cellular caller -> rotary handset



Startup: set VR1 so wiper-to-ground is approximately 0 Ω, then raise slowly during a call.

R1 protects/isolates the BTL output; AG1171 VIN is AC-coupled and level-adjustable.

B. GPIO36 DIAL / BUSY / REORDER TONE INJECTION



Firmware starting settings

PWM carrier: 31 kHz or higher

Tone: 350 Hz + 440 Hz

Stop immediately at first dial pulse

R2/C3 are a first-order PWM filter.

Common ground: LilyGO GND, AG1171 GNDPWR, VR1 ground, and C3 ground.

Rev A - receive section is the recommended first experiment

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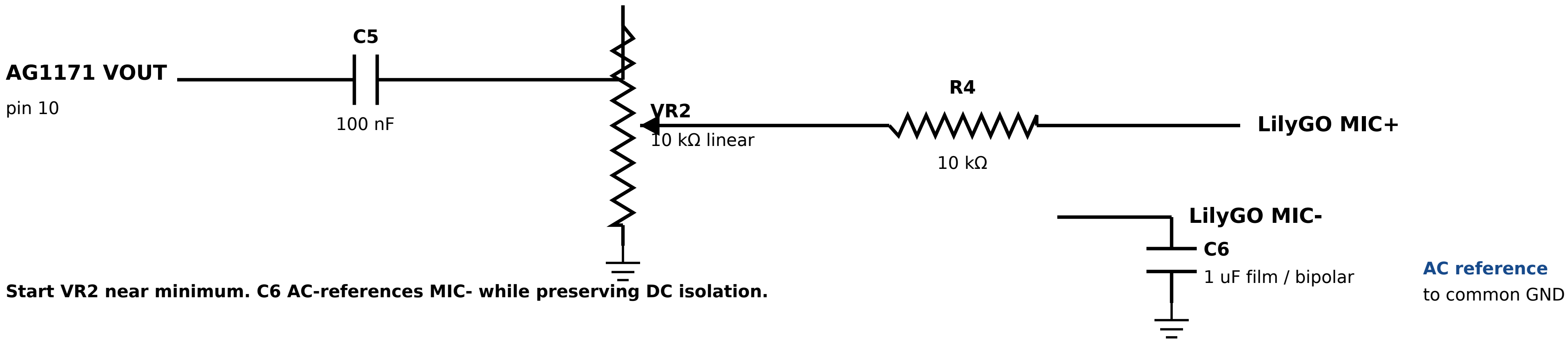
Sheet 2 - Passive transmit test path and BOM

PROTOTYPE - BENCH VALIDATION REQUIRED

C. TRANSMIT: rotary handset -> cellular caller (bench-test circuit)

Measured idle bias: MIC+ ≈ 0 V, MIC- ≈ 0 V, and MIC+ to MIC- ≈ 0 V.



Start VR2 near minimum. C6 AC-references MIC- while preserving DC isolation.

D. RECORDED LILYGO AUDIO DC READINGS

Measured powered and idle, with the audio board disconnected:

- SPEK+ to GND: approximately 1.8 V
- SPEK- to GND: approximately 1.8 V
- SPEK+ to SPEK-: only a few mV
- MIC+ to GND: approximately 0 V
- MIC- to GND: approximately 0 V
- MIC+ to MIC-: approximately 0 V

SPEK- remains floating. MIC- is AC-grounded only through C6.

E. STARTING BOM / BUILD NOTES

- VR1, VR2: 10 kΩ linear trimmers
- R1, R3, R4: 10 kΩ; R2: 3.3 kΩ
- C1: 1 uF film or bipolar
- C2: 10 nF; C3: 22 nF; C4: 220 nF
- C5: 100 nF; C6: 1 uF film or bipolar
- J1: 2-pin SPEK connector; pads for MIC+, MIC-, GND, GPIO36
- Keep the cellular antenna and boost/switching inductors away from audio wiring.

This is the starting transformerless bench circuit. Validate levels and noise before permanent construction.

No transformers / no op-amps